

REFINING RESEARCH QUESTIONS & DELIVERABLES

Group/Individual Check-In

It is good practice to reflect! Before moving forward with your project deliverables, take a moment as a group or independently to check in on your progress:

- **Pacing:** *Do you need to slow down or speed up?*
- **Comfort level:** *How comfortable do you currently feel with your project?*
- **Team Communication:** *For groups, have you been actively communicating with your team members?*
- **Identify Gaps:** *Reflect on your current research question. What is missing, and how can you fill that gap?*



Elements of a Refined Research Question

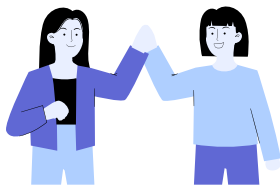
When you take out your current research question, evaluate its specificity. A strong, refined question should move beyond a simple yes/no answer and include:

- **Population Impacted:** *Clearly define the specific population that is affected by your study.*
- **Treatment Groups:** *Specify your exact treatment groups or the prototype you are analyzing.*
- **Broader Implications:** *Instead of just asking if Molecule A can replace Molecule B, add an extension. For example: "If so, what implications exist for the cost and longevity of the final product?"*
- ***Value*:** *Always explicitly consider what your specific study brings to the table.*



Red Flags to Watch Out For:

- **Too Easily Answered:** *You can look it up and find the answer online, or a single article alone already answers it.*
- **Opinion-Based:** *The question is built on opinions that you feel sound right, rather than hard facts.*
- **Already Disproven:** *The premise has already been disproven by existing science (if so, find a new route!).*
- **No Direct Sources:** *There are no direct or close sources available for what you want to study, making it highly difficult to find supporting evidence for your methods*



Feedback + Deliverables

Give & Receive Feedback! Getting a new set of eyes and adapting to constructive criticism will help deepen your project

Final step - Deliverable: *Set a goal for what project you want to create for the end of this program. This ranges from an educational website to a research paper! Be creative and have fun*